Daily activity does not predict next-night sleep quality

An n-of-1 wearable study across six consensus sleep indicators

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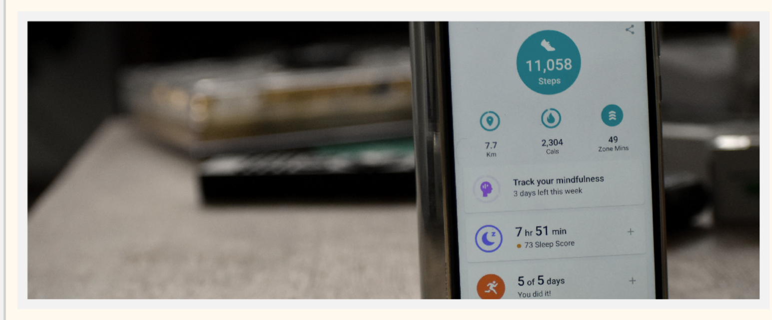

INTRODUCTION

Sleep quality lacks a definitional consensus. A National Sleep Foundation panel endorsed the continuity measures — efficiency, wake after sleep onset (WASO) and onset latency — but reached no consensus on sleep architecture.

The two-process model predicts daytime behaviour should shape the sleep that follows, and meta-analysis finds small benefits of acute exercise. But those effects are largely between persons. A within-person study found activity and sleep were related only once perceived stress was accounted for, not as a direct daily effect on its own.

AIM

Can the previous day's step count and energy expenditure predict six sleep quality indicators on the following night, in a single adult over 151 consecutive nights?

METHOD

151 nights, one adult, consumer sleep-tracking device, Nov 2025 – May 2026
Noon-to-noon nights; longest episode = main sleep, rest are naps
Outcomes: efficiency, total sleep time, WASO, onset latency, REM %, deep %
Predictors: steps, kcal, previous-night efficiency and bedtime, 7-night bedtime SD, weekend
Nothing inside the outcome window used as a predictor
Four model families; three nested predictor sets
Blocked forward-chaining cross-validation, 5 folds
Every result compared against an uninformative baseline

RESULTS

0 / 36	0.04	< 0
predictor–outcome correlations significant	lag-1 autocorrelation of sleep efficiency	cross-validated R^2 on all six outcomes

No model beat its baseline on any outcome. The mean baseline achieved the best R^2 on five of six outcomes, and every AUC confidence interval included 0.50.

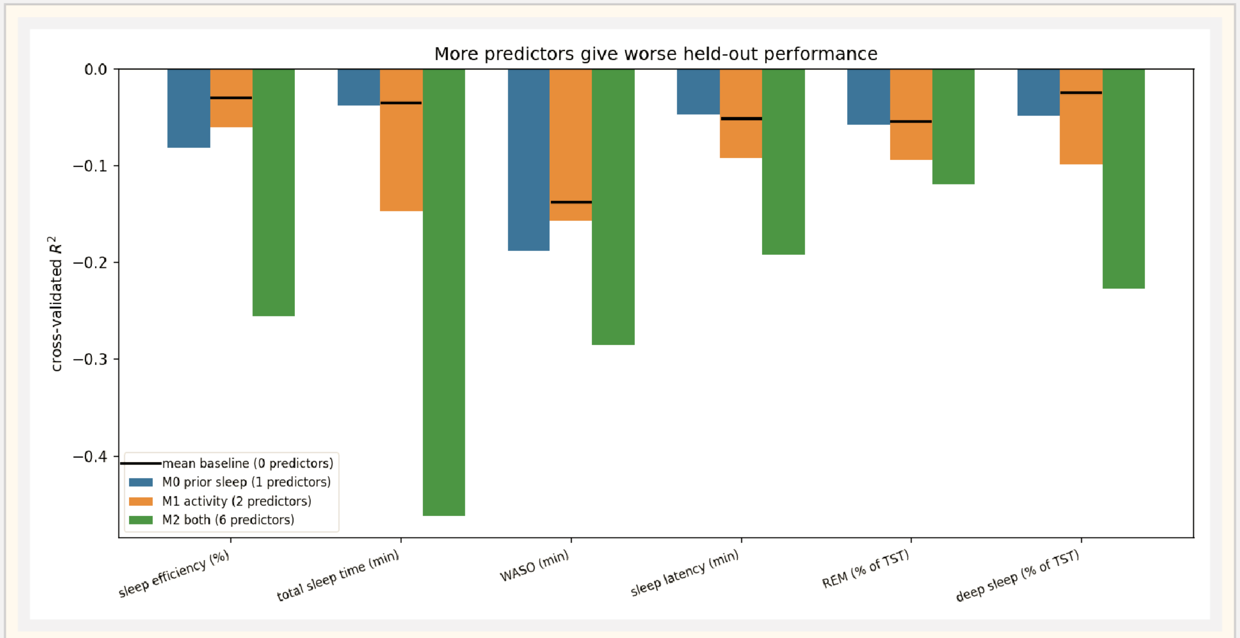

Fig. 1 Performance falls as predictors are added. Horizontal markers show the zero-predictor baseline — it outperforms every fitted model.

POSITIVE CONTROL

A null result invites the objection that the pipeline is broken. Applying it unchanged to two predictors that determine the outcome by construction — time asleep and time in bed, whose ratio defines efficiency — recovered AUC 0.873, against 0.410 on the real predictors. Same folds, same metrics, same code. The pipeline detects signal when signal is present, so the absence is a property of the data.

A FINDING EFFICIENCY HIDES

Two consensus continuity measures changed significantly over six months — in opposite directions. Sleep efficiency, the most widely used single metric, shows nothing.

OUTCOME	1ST	3RD	P
WASO (min)	33.5	51.3	0.001
Latency (min)	28.8	19.7	0.004
REM (%)	26.4	29.4	0.010
Efficiency (%)	86.3	85.7	0.72

Table 1 Outcome means by third of the period. WASO rose 53% and latency fell 32%; the changes cancel in the efficiency ratio.

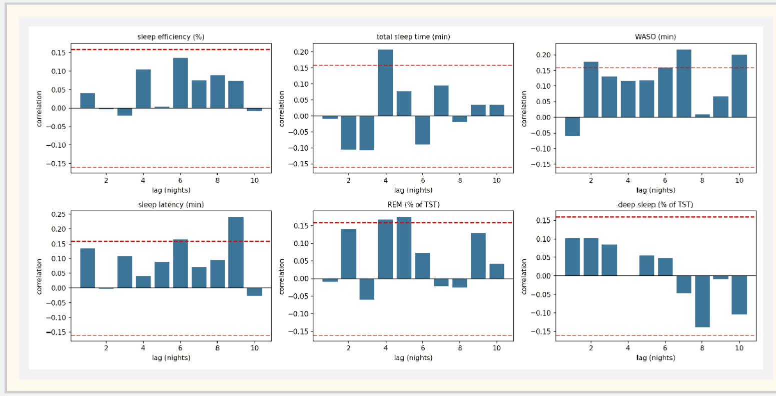

Fig. 2 Lag-1 autocorrelation is near zero for every outcome; isolated crossings at longer lags are consistent with chance across 60 tests, not persistence.

CONCLUSIONS

Daily step and calorie totals did not predict any of six sleep indicators in this individual.
Consistent with within-person literature. The study reaching AUC 0.94 used within-day activity sequences, not daily totals.
Without real signal, model capacity is purely harmful — 6 predictors < 2 predictors < none.
Reporting an uninformative baseline is essential: 59% accuracy looks usable until the majority baseline is 66%.
Efficiency alone concealed real change — report continuity measures separately.

Key references: Mendonça et al., IEEE Access 7, 24527 (2019) · Ohayon et al., Sleep Health 3, 6 (2017) · Borbély, Hum. Neurobiol. 1, 195 (1982) · Kredlow et al., J. Behav. Med. 38, 427 (2015) · Sathyanarayana et al., JMIR Mhealth 4, e125 (2016) · Collins et al., BMJ 385, e078378 (2024). Reported per TRIPOD+AI.